

CCoT-Emo

*Plug-and-Play Emotion Graphs for Compositional Prompting
in Zero-Shot Speech Emotion Recognition*

Jiacheng Shi, Hongfei Du, Y. Alicia Hong & Ye Gao

IEEE ICASSP 2026 · arXiv:2509.25458

College of William & Mary, with George Mason University

Structured prompting and low-resource

Why this paper matters



Better emotion recognition with no gradient updates at all

0

parameters updated —
no fine-tuning anywhere

+9.1

points average gain on
Qwen2-Audio across five sets

7

acoustic features, discretised
into low / normal / high

3

audio LLMs it plugs into
without modification

● The claim

Audio language models underrepresent paralinguistic cues. Rather than teaching them to attend to pitch and jitter through fine-tuning, extract those cues externally, serialise them into a structured Emotion Graph alongside textual sentiment and cross-modal relations, and paste the graph into the prompt. The model then reasons over an explicit symbolic trace instead of an implicit one.

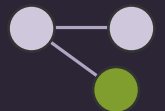
● The contrast with last session

EAA reached 0.687 on MELD by training a fusion module and a LoRA adaptor on one H100. CCoT-Emo reaches comparable gains by changing the text in the prompt — and the two papers share an author. Reading them together is a study in what you buy with weights versus with structure.

1

Unstructured reasoning is not enough

Why plain chain-of-thought fails here



The diagnosis



Two failures, one in the models and one in the prompting fix

● The models underrepresent prosody

Audio LLMs often focus on linguistic semantics and underrepresent paralinguistic cues such as pitch, speech rate, volume, jitter, shimmer, intensity and articulation rate.

We have evidence for this from the whole series: Pengi at chance on CREMA-D, SALMONN at 0.69 against an 0.81 specialist, Qwen-Audio with under 1,000 hours of emotion data in a mix exceeding a hundred thousand.

● And chain-of-thought does not fix it

Prompting-based methods ask the model to reason step by step through emotional inference, surfacing latent paralinguistic reasoning as natural-language explanations.

The paper's objection: because these traces are unstructured, they are vulnerable to hallucination and error propagation, and their interpretability remains largely implicit.

● The proposal

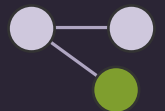
Replace the unstructured reasoning trace with a symbolic one. Extract the acoustic facts deterministically, state them explicitly, and let the model reason over the statement rather than over the signal.

The inspiration is named: compositional chain-of-thought prompting from the vision–language literature, which uses structured representations of objects, attributes and relations.

2

Building an Emotion Graph

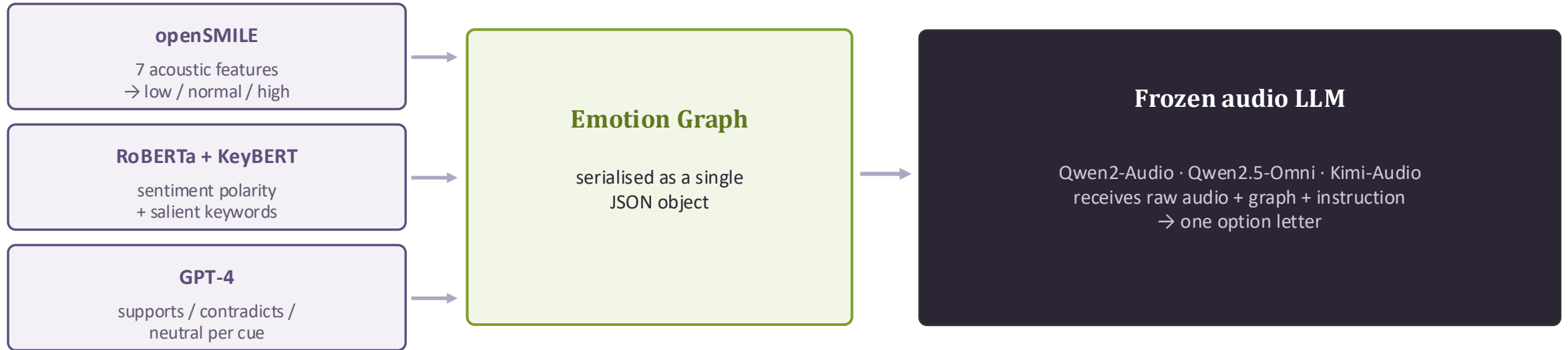
Three sources of structure, one JSON object



The two-stage pipeline



Stage 1 builds the graph; stage 2 prompts with it



- All three graph components are computed independently and then serialised into a unified JSON structure.
- Nothing about the graph construction is learned or tuned on emotion labels — the extractors are off the shelf.
- The audio LLM still receives the raw audio. The graph is additional conditioning, not a replacement for listening.
- Because the graph is text, the method applies to any instruction-following audio LLM without architectural change — the 'plug-and-play' claim.

The two-stage pipeline

Stage 1 builds the graph; stage 2 prompts with it

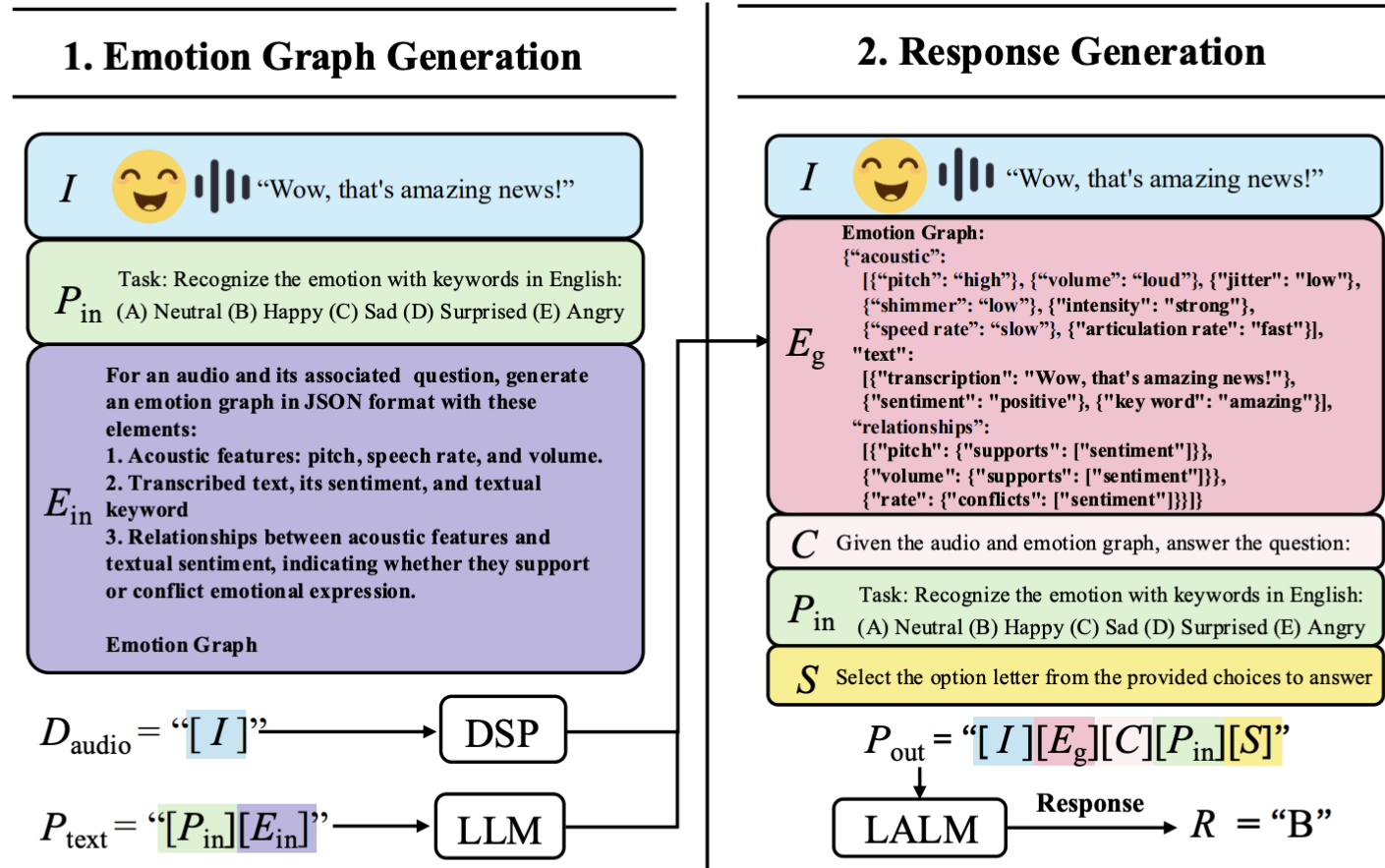


Fig. 1. Overview of full CCoT-Emo. Stage 1 builds emotion graph (EG) by extracting acoustic features via digital signal processing (DSP) [14] and deriving textual sentiment, keywords, and cross-modal relationships using an LLM. Stage 2 prompts the LALM with the audio, EG, and task instruction to predict the emotion label. Components unique to CCoT-Emo are in **bold**.

The acoustic attributes



Seven classical measures, deliberately discretised

Pitch

fundamental frequency

Speech rate

syllables per unit time

Volume

loudness

Jitter

cycle-to-cycle frequency variation

Shimmer

cycle-to-cycle amplitude variation

Intensity

signal energy

Articulation rate

speaking rate excluding pauses

Extraction

Computed with openSMILE over the full utterance — the same toolkit behind eGeMAPS, which we met in the MSP-Podcast session as the retrieval feature set.

Discretisation

Each feature is binned into low, normal or high based on empirical value distributions computed over the corpus — so the thresholds are corpus-relative, not absolute.

The discretisation is the design decision

A continuous pitch value in hertz means nothing to a language model. The categorical label 'high' is a token it can reason about. This is the step that turns a measurement into something compositional — and section 4 shows what happens when you skip it.

The textual and cross-modal attributes



Two more models in the pipeline

● Textual attributes

A RoBERTa-based sentiment classifier predicts sentence-level polarity as positive, negative or neutral from the transcript.

KeyBERT additionally extracts emotionally salient keywords by semantic similarity.

For IEMOCAP and MELD the transcripts are the corpora's own; for ESD and MERBench, which lack transcripts, Whisper is used.

● Cross-modal relationships

GPT-4 is prompted with the acoustic cues and the textual sentiment and asked to determine whether each cue supports, contradicts, or is neutral with respect to the sentiment.

This is the part that captures the interaction the paper is after: a high pitch supporting a positive sentiment reads differently from a high pitch contradicting one.

● Count the models in the pipeline

openSMILE, a RoBERTa sentiment classifier, KeyBERT, GPT-4, sometimes Whisper — and then the audio LLM itself. Six components for one prediction.

None of them is fine-tuned on the target task, so 'training-free' is accurate. But the RoBERTa classifier was supervised on sentiment, and GPT-4 is doing a reasoning step before the 'zero-shot' model reasons. Hold that until section 5.

Stage 2: the prompt



Five concatenated elements

I

the raw audio input

E_g

the Emotion Graph as JSON

C

"Use the audio and emotion graph as context and answer the following question."

P_in

"Task: Recognize the emotion with keywords in English: (A) Neutral (B) Happy (C) Sad (D) Surprised (E) Angry."

S

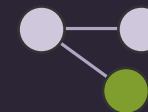
"Select the option letter from the provided choices to answer."

The forced-choice output instruction is borrowed from LLaVA and exists to reduce output ambiguity — which, from the Pengi session, we know is worth several points on its own.

3

Results

Five benchmarks, three audio LLMs



Setup



Four corpora, five test sets, and protocol choices worth noting

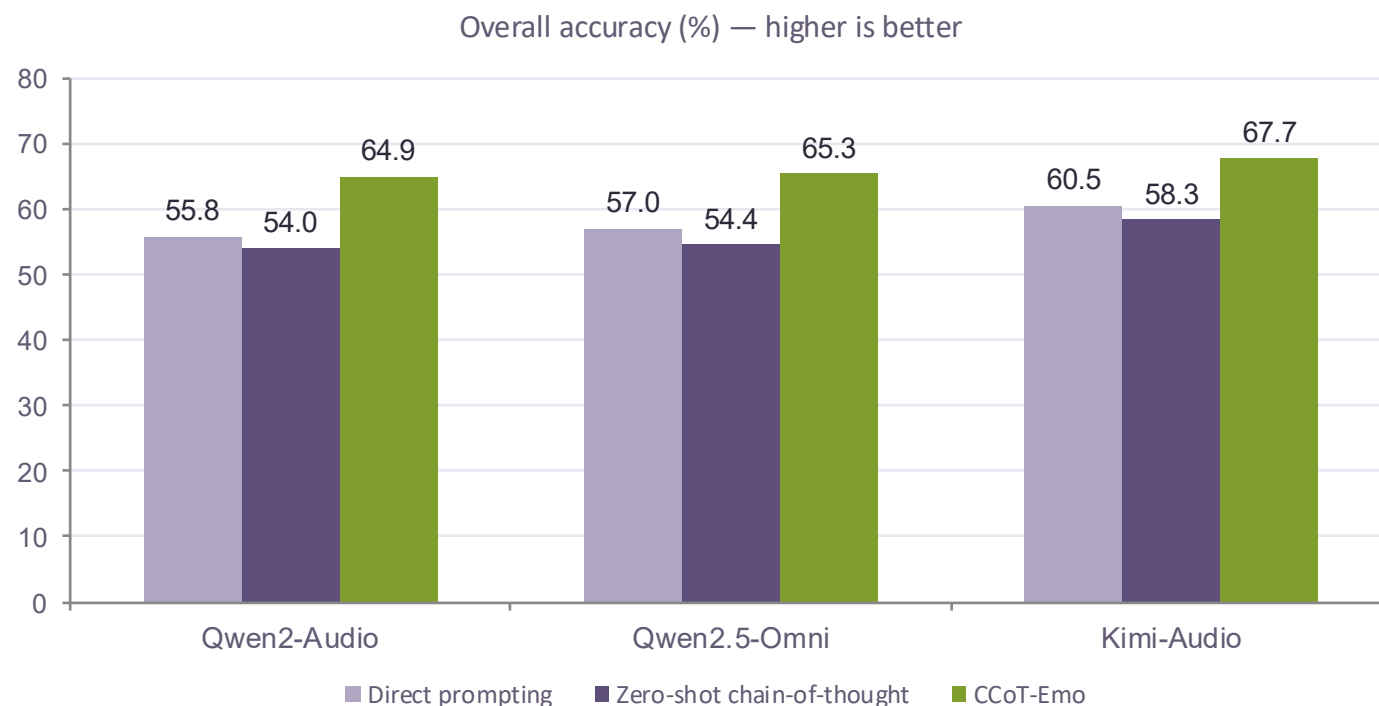
Benchmark	Content	Protocol	Transcripts
IEMOCAP	5,531 English utterances, five dyadic sessions	Excited merged into happy, 5-fold leave-one-session-out	Corpus
MELD	8,851 English utterances from Friends	Disgust and fear removed — five classes, not seven	Corpus
ESD	33,443 utterances, English and Chinese, five emotions	Diverse speakers	Whisper ASR
MERBench test1	350 Chinese utterances from film and television	Worried label excluded	Whisper ASR
MERBench test2	350 Chinese utterances from film and television	Worried label excluded	Whisper ASR

- Since the method is training-free, evaluation runs directly on the test sets — no training split is consumed.
- The protocol choices follow BLSP-Emo, which is also the strongest prior baseline, so the comparison is like for like.
- Three of the five sets are Chinese or bilingual, which is a genuine breadth advantage over most of the papers we have read.
- On ESD and MERBench the textual half of the emotion graph is built from ASR output, so transcription errors propagate into the structure.

Main results



Average accuracy across the five benchmarks



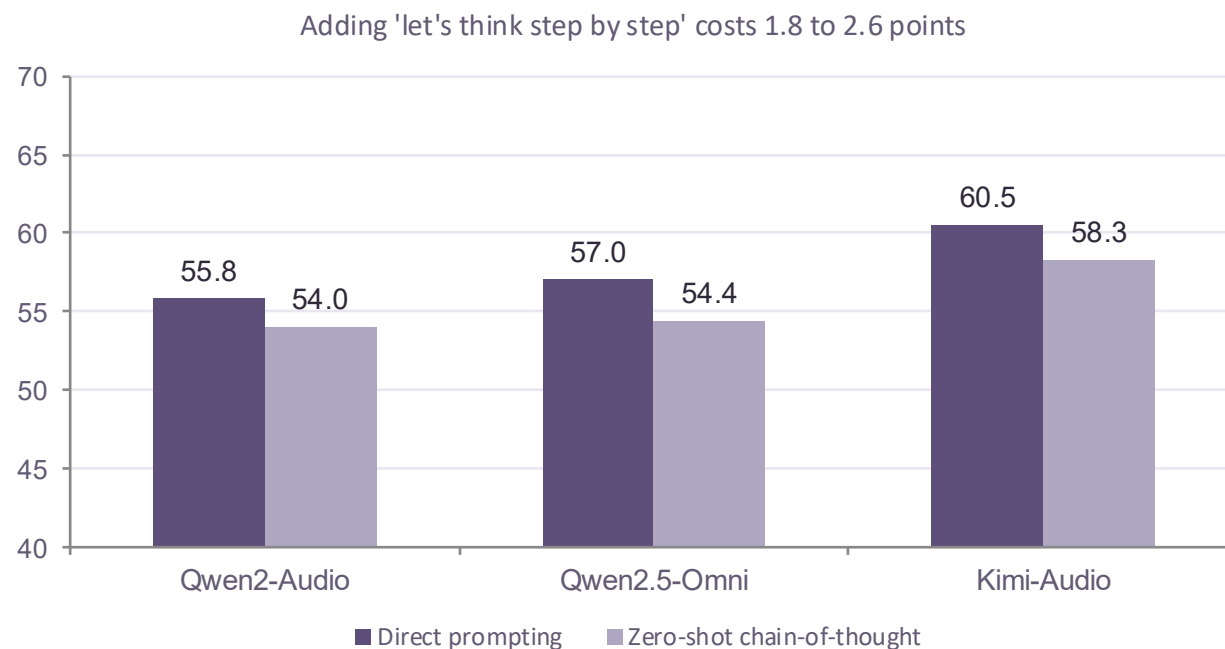
- Gains of +9.1, +8.3 and +7.2 points over direct prompting, consistent across three architectures.
- Kimi-Audio with CCoT-Emo reaches 67.7, beating the previous best audio LLM, BLSP-Emo, by 3.7 points.
- The largest single-dataset gain is ESD for Qwen2-Audio: 57.1 to 71.6, a 14.5-point jump.
- Gains hold on the domain-shifted Chinese MERBench sets — +8.5 and +7.5 for Kimi-Audio.

The consistency across three different audio LLMs is the strongest part of the result — it supports the plug-and-play claim better than any single number does.

Chain-of-thought makes it worse



The finding worth pausing on



- ZS-CoT underperforms direct prompting on all three models and on nearly every dataset.
- The paper's explanation: irrelevant intermediate reasoning disrupts emotion inference, particularly in datasets with long-range context.
- So the gain is not 'reasoning helps'. It is specifically structured reasoning that helps, and unstructured reasoning actively hurts.
- That makes the comparison against ZS-CoT the more informative baseline — CCoT-Emo beats it by 9.4 points on Kimi-Audio.

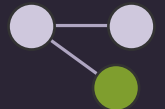
● Connect this to SALMONN

SALMONN's task over-fitting analysis showed that a model asked to tell a story would produce a transcript, because its prior over responses was concentrated on transcription. Here, a model asked to reason freely about emotion produces reasoning that drags the answer away from the right one.

4

Ablations

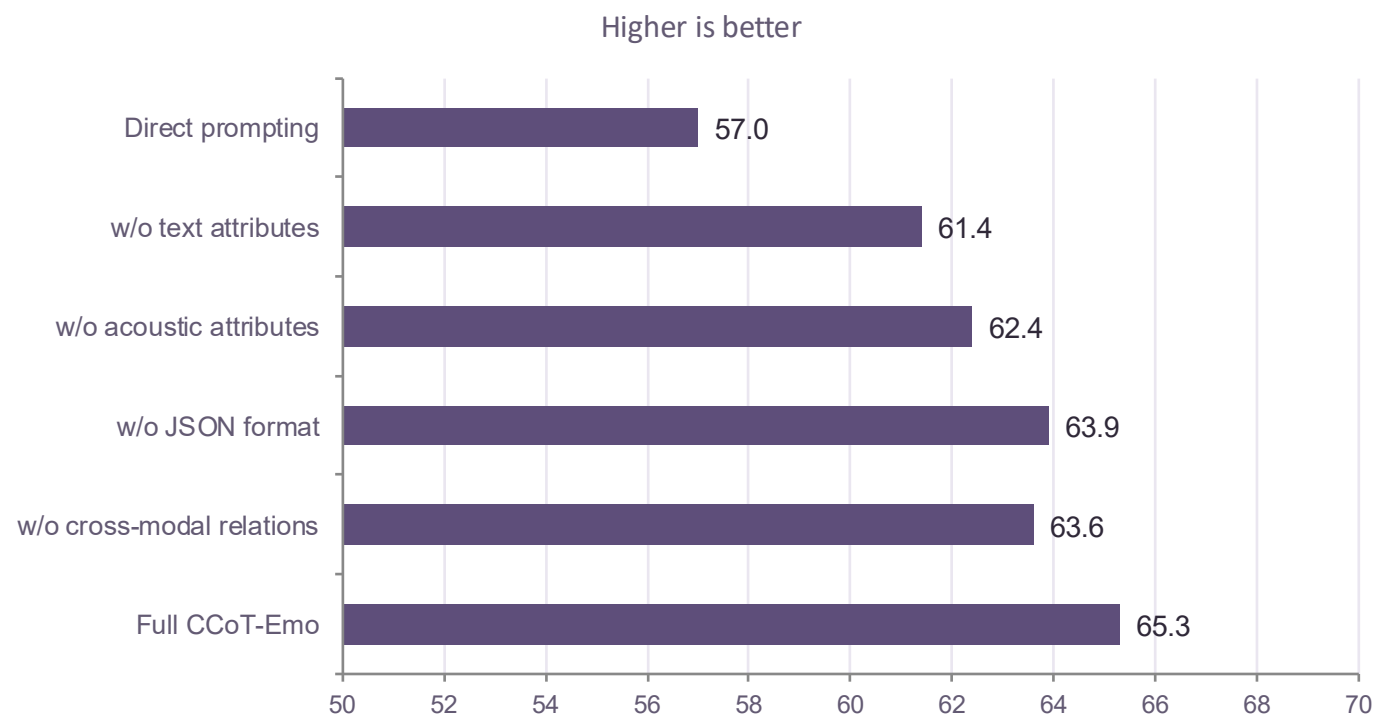
Where the gain comes from



Component ablation



Qwen2.5-Omni-7B, removing one piece at a time



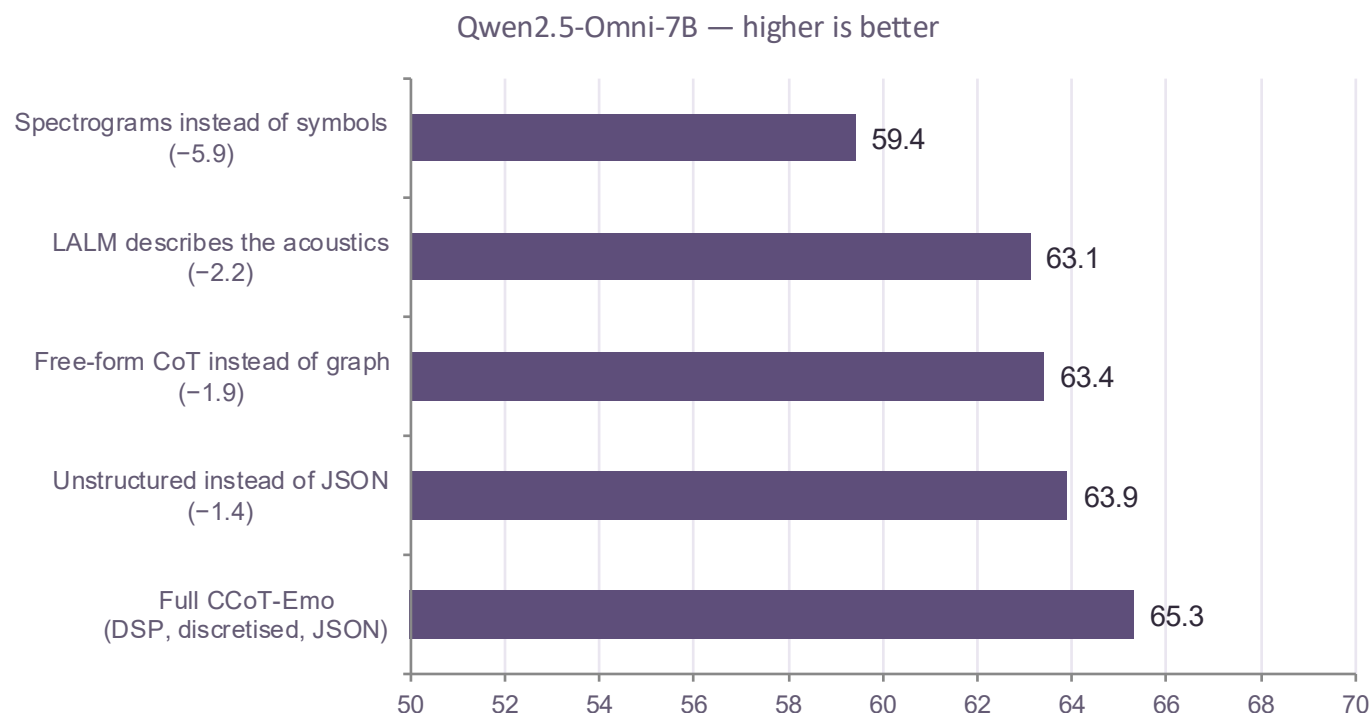
- Removing textual attributes — sentiment and keywords — costs the most: 3.9 points.
- Removing acoustic attributes costs 2.9; removing cross-modal relations costs 1.7.
- So the single largest contributor is the output of a supervised RoBERTa sentiment classifier reading the transcript.
- That is the same pattern we found in EAA last session, where the transcript carried 14.4 of the 16.4 points in the context ablation.

Two papers, two mechanisms, same finding: on these benchmarks the linguistic channel does more work than the acoustic one, even in methods designed to foreground prosody.

Symbolic beats continuous, and DSP beats the model



Three ablations about representation



- Asking the audio LLM to describe its own prosody instead of using openSMILE costs 2.2 points — the paper reports such descriptions are vague, e.g. 'the tone is expressive'.
- Replacing the symbolic features with spectrograms costs 5.9 points, the largest drop of any ablation.
- The authors' reading: continuous and uninterpretable representations lack the structural abstraction required for zero-shot symbolic reasoning.
- Even the JSON formatting matters — 1.4 points for serialisation alone.

The spectrogram result is the paper's strongest evidence for its thesis: the gain is not from adding acoustic information, it is from adding it in a form the model can reason over.

Two scaling ablations



Model size and graph size

● Model size

Qwen2.5-Omni 3B: 46.6 direct, 50.9 with CCoT-Emo.

Qwen2.5-Omni 7B: 57.0 direct, 65.3 with CCoT-Emo.

The paper's conclusion: larger models better exploit structured intermediate representations for emotion reasoning.

● Graph size

Token limits swept from 128 to 1024, with 256 as the default.

256 is best. Shorter costs 0.4 points; 512 costs 0.7; 1024 costs 0.9.

Compact, well-balanced emotion graphs offer the best guidance — more structure is not better.

● Both echo findings from earlier sessions

The model-size result is Lester's power of scale again: a bigger frozen model extracts more from the same conditioning signal.

The graph-size result is the prompt-length curve again: sharp gains to a point, then diminishing and finally negative returns. Lester found the plateau at 20 tokens, CLEP-DG at 8, and here the optimum is 256 — different units, same shape.

Two arithmetic notes



An exercise in reading tables carefully

● The Qwen2-Audio gain

The abstract and Table 1 both report +9.1 points for Qwen2-Audio: 55.8 direct to 64.9 with CCoT-Emo.

The results section states that CCoT-Emo enhances Qwen2-Audio and Qwen2.5-Omni by +6.1% and +8.3% respectively.

The +8.3 matches; the +6.1 does not match either the table or the abstract.

● The 3B gain

Table 2 gives Qwen2.5-Omni-3B at 46.6 and 3B with CCoT-Emo at 50.9 — a difference of 4.3 points.

The ablation text says the 3B variant improves SER accuracy by 3.9%.

Neither discrepancy changes a conclusion.

● Why raise this in a seminar

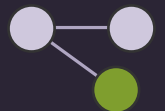
We have found one of these in almost every paper this term — wav2vec 2.0's 5.2/8.6, AffectNet's RMSE, MSP-Podcast's kappa-versus-alpha labelling, EAA's Table 3 averages.

The habit generalises: when a paper states an improvement in prose, check it against the table it came from. These are the kind of numbers that propagate into secondary literature if nobody looks.

5

Appraisal

What 'zero-shot' and 'plug-and-play' mean here



Interrogating the two labels



Both claims are true, and both need qualifying

● Zero-shot — for the audio LLM

Accurate. No parameter of Qwen2-Audio, Qwen2.5-Omni or Kimi-Audio is updated, and no emotion-labelled training data is consumed by any component of the pipeline.

● Zero-shot — for the system

Qualified. The RoBERTa sentiment classifier is supervised on sentiment, and its output is the single largest contributor in the component ablation. No emotion labels, but not label-free.

● Plug-and-play — for integration

Accurate. The graph is text, so it drops into any instruction-following audio LLM, and the paper demonstrates this across three architectures.

● Plug-and-play — for deployment

Qualified. Each prediction requires openSMILE, a sentiment classifier, KeyBERT, a GPT-4 call, sometimes Whisper, and then the audio LLM. Latency and cost are not reported.

Critical appraisal



What the design buys, and the open threads

● Strengths

Seven ablation conditions in a four-page paper — unusually thorough for the format, and they test the thesis rather than just the system.

The spectrogram and LALM-describes-prosody ablations are the right controls: they isolate the form of the information from its presence.

Consistent gains across three architectures, which supports the plug-and-play claim better than any single number.

Genuine multilingual breadth — three of five test sets are Chinese or bilingual.

An interpretable intermediate representation that a human can read and check, which almost nothing else in this series offers.

The negative ZS-CoT result is reported rather than buried.

● Open threads

No graph-only control: we cannot tell whether the audio contributes anything once the graph is present.

The largest component is a supervised text sentiment classifier, which the paper does not remark on.

Accuracy only, on corpora known to be imbalanced — no unweighted accuracy or per-class breakdown.

No cost or latency reporting for a six-component pipeline with a GPT-4 call per utterance.

Acoustic discretisation thresholds are computed over the corpus, which complicates single-utterance deployment.

No seeds or variance; generation from LLMs is stochastic and single runs are reported.

Take-aways



Four things to carry into the second half

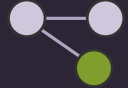
- 1 Structure beats narration**

Free-form chain-of-thought is reliably negative here; a symbolic graph is reliably positive. The gain is from constraining the intermediate representation, not from adding one.
- 2 Form matters as much as content**

The same acoustic information as spectrograms costs 5.9 points against discretised symbols. Representation is not a packaging decision.
- 3 Measure outside the model**

Asking the audio LLM to describe its own prosody costs 2.2 points. openSMILE cannot hallucinate a pitch value.
- 4 The text channel keeps dominating**

Textual attributes are the largest single contributor here, as the transcript was in EAA. Two mechanisms, same conclusion about these benchmarks.



Questions for discussion

1

The model receives both the audio and the graph. What would a graph-only control return, and what would you conclude if it matched the full system?

2

Zero-shot chain-of-thought is reliably negative across three models. Why should narration hurt here when it helps on arithmetic and commonsense reasoning — and what does that say about what kind of task emotion recognition is?

3

Textual attributes contribute more than acoustic ones, and they come from a supervised sentiment classifier. Is this better described as zero-shot emotion recognition or as cross-domain transfer from text sentiment?

4

Replacing symbols with spectrograms costs 5.9 points. Is the lesson about audio LLMs specifically, or about what any language model can reason over?

5

EAA fine-tuned on one H100; CCoT-Emo calls GPT-4 per utterance. Which is actually cheaper, and under what deployment assumptions?